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RESEARCH ARTICLE

Unified Object Detector With Dynamic Switch Mechanism Based on Deep Learning Advances: A Paradigm-Aware Architecture for Adaptive Detection

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ABSTRACT Object detection research has evolved through different architectural paradigms, including two-stage region-based models, one-stage efficiency-focused detectors, and more recent transformer methods, where each performs well in a specific domain but none is optimal overall. This fragmentation requires static architectural decisions by developers and often leads to suboptimal performance when application requirements change. In this work, we address these limitations through two complementary and novel contributions. First, we conduct a systematic bibliometric analysis of the object detection literature from 2014–2025 to derive an empirical paradigm–domain relationship matrix, enabling the identification of the most influential detection paradigms together with their dominant application domains. This provides, for the first time, a data-driven mapping between detection paradigms and domain-specific requirements. Second, we propose a novel Unified Object Detector with Dynamic Switch Mechanism (UOD-DSM), which operationalizes the insights obtained from the bibliometric analysis into an adaptive detection architecture. The UOD-DSM integrates multiple detection paradigms within a unified framework and employs a context-aware switching mechanism to dynamically select the most suitable paradigm at runtime according to operational constraints and input characteristics. Experimental validation demonstrates that the proposed framework offers improved flexibility and robust performance across diverse applications, including real-time video analysis and precision-demanding medical imaging, while outperforming conventional static detectors in cross-domain evaluation. These results demonstrate the potential of the proposed approach to advance object detection toward fully adaptive vision systems capable of aligning architectural strategies with varying operational contexts.

INDEX TERMS Deep learning, object detection, dynamic networks, paradigm switching, bibliometric analysis, adaptive computer vision, neural architecture selection.

I. INTRODUCTION

Object detection is a cornerstone of computer vision and enables machines to localize and classify objects in digital images as well as video. Its applications are used everywhere and have important value, including autonomous driving [1], robotic perception [2], medical image diagnosis [3], and

intelligent video surveillance [4]. The field functions in a multidisciplinary environment, where ideas are drawn from computer science, electrical engineering, and decision theory to produce domain-specialist algorithmic ad hoc solutions. The performance of any object detection system is typically measured along two, sometimes conflicting, directions: **accuracy**, which is usually presented by popular metrics including mean Average Precision (mAP), and **efficiency**, such as frames per second (FPS) or inference latency.

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The architectural evolution of object detectors has been marked by distinct, dominant paradigms, each addressing core performance trade-offs. The era of **two-stage, region-based detectors** (circa 2014–2018), pioneered by region-based convolutional neural network (R-CNN) [5] and significantly advanced by Faster R-CNN [6], established strong accuracy benchmarks through a propose-then-classify pipeline, albeit at high computational cost. This was followed by the rise of **one-stage, efficiency-optimized models** (circa 2019–2021), such as the YOLO (You Only Look Once) series [7], [8] and SSD (Single Shot MultiBox Detector) [9], which achieved real-time performance by unifying proposal generation and classification into a single network pass, often with a modest compromise on accuracy, particularly for small objects. Most recently, **Transformer-based and hybrid architectures** (circa 2022–present) have emerged, with Vision Transformers (ViTs) and Detection Transformers (DETR) [10] introducing global contextual modeling via self-attention mechanisms. Subsequent hybrids like ConvNeXt [11] and YOLOS [12] blend CNN inductive biases with Transformer strengths, pushing state-of-the-art (SOTA) accuracy frontiers, frequently with substantial computational demands.

This rapid paradigm-focused evolution has resulted in a fragmented and complex research landscape. A long-standing and central problem is the lack of a one-size-fits-all detector architecture that performs optimally across all application domains and their trade-off operational requirements [13]. As a result, practitioners and system designers are left with ad hoc decisions made a priori: selecting one-stage detectors for latency-sensitive applications or more powerful two-stage or Transformer-based models for accuracy-critical tasks [14]. Such a static commitment inevitably leads to performance degradation when deployment conditions are diverse or variable, particularly in scenarios that require balancing speed and accuracy. As discussed in detail in [15], the “best” detector is highly dependent on contextual factors such as object scale, scene complexity, and hardware constraints.

Research efforts to navigate these trade-offs have focused on innovations within individual paradigms. Within the two-stage family, Cascade R-CNN [16] improved precision via multi-stage refinement, while Mask R-CNN [17] extended the framework to instance segmentation. For one-stage detectors, architectural refinements like Feature Pyramid Networks (FPN) [18] and its variants (e.g., bi-directional feature pyramid network (BiFPN) [19]) enhanced multi-scale detection capability, and novel loss functions like Focal Loss [14] mitigated class imbalance. The Transformer paradigm has seen efforts to improve efficiency and convergence, exemplified by Deformable DETR [20] and real-time variants like Real-Time DETECTION TRANSformer (RT-DETR) [21]. Parallel research threads in Neural Architecture Search (NAS) [22] and model compression (e.g., pruning [23], quantization [24]) seek to optimize

designs automatically or reduce their footprint. A promising, yet distinct, direction involves dynamic networks that employ conditional computation, such as dynamic convolution [25] or early exiting mechanisms [26]. However, these approaches primarily adapt the computational intensity *within* a fixed architectural paradigm rather than selecting *between* fundamentally different detection paradigms.

Despite the rapid architectural progress across these paradigms, the literature largely evaluates detection models within isolated architectural families rather than examining their broader suitability across heterogeneous application domains. Most comparative studies focus on benchmarking detectors under fixed datasets or standardized evaluation settings, emphasizing improvements in accuracy, efficiency, or convergence behavior. While these efforts have significantly advanced individual architectures, they rarely provide a systematic perspective on how different paradigms align with specific operational requirements such as speed, contextual complexity, or precision-critical requirements. Consequently, model selection in practical deployments is often guided by heuristic judgment or domain experience rather than by a structured empirical understanding of paradigm–domain suitability. This lack of a unified framework, such as the UOD-DSM, highlights the need for research that not only characterizes the evolving landscape of object detection paradigms but also motivates the development of adaptive architectural strategies capable of responding to varying contextual requirements.

Therefore, a critical gap remains: **there exists no systematic, data-driven framework to map high-level detection paradigms to the application domains where they are most effective, and no unified architecture that can dynamically execute this mapping at runtime.** Current solutions are either static or adaptively efficient only within a narrow architectural family, failing to leverage the full spectrum of strengths offered by the diverse historical evolution of object detection.

To bridge this gap, this paper is structured around two complementary aims:

- **Aim 1:** To identify and map the most influential object detection paradigms to their optimal application domains using a systematic, data-driven bibliometric analysis.
- **Aim 2:** To develop a novel UOD-DSM that operationalizes the insights from Aim 1 into an adaptive architecture capable of selecting the optimal detection paradigm at runtime.

Our work makes the following primary contributions:

- **C1: A Data-Driven Bibliometric Analysis and Paradigm-Domain Map.** We conduct a systematic bibliometric analysis of the object detection literature from 2014 to 2024. Using scientometric tools like VOSviewer, we analyze citation networks and keyword co-occurrence to identify seminal works, trace paradigm evolution, and uncover natural research clusters. This

analysis yields an empirical **paradigm-domain relationship matrix**, providing a quantitative foundation for optimal paradigm selection.

- **C2: A Novel UOD-DSM.** We propose a novel unified detector architecture that encapsulates multiple core detection paradigms (e.g., a one-stage sub-network, a two-stage sub-network, a lightweight Transformer sub-network) within a single framework. Its core innovation is a **DSM**. The DSM translates the insights from C1 into executable runtime decision rules. By processing contextual inputs—such as target latency budget, required confidence threshold, and inferred scene complexity—the DSM activates the most appropriate sub-paradigm for the current input, enabling a single model to excel across diverse requirements.

The new class of UOD-DSM goes beyond the static design and opens up a way toward truly adaptive and efficient vision systems. The remaining sections of this paper are arranged around the following research framework and empirical findings. In section II, we summarize the related work about object detection paradigms and adaptive networks. Section III provides an in-depth description of the methodology integration, split into the bibliometric analysis framework (C1) in III-A and the conceptual architecture of the UOD-DSM (C2) in III-B. Section IV is a critical analysis and interpretation of the empirical findings broken down according to results for Aim 1 (IV-A) and Results for Aim 2 (IV-B), with focused discussion comments in section IV-C, and specific explanations in section IV-D. Section V evaluates the theoretical and practical constraints relative to Aim 1 (V-A) and Aim 2 (V-B), and further articulates the foundational research assumptions underpinning the proposed framework (V-C). Finally, Section VI synthesizes the research findings and delineates prospective trajectories for future research.

II. RELATED WORKS

The evolution of deep learning-based object detection has progressed through several distinct phases, each addressing fundamental challenges in localization, scale invariance, and computational efficiency. This review organizes the key developments chronologically, highlighting the paradigm shifts and technical innovations that have defined each period.

A. FOUNDATIONAL CONCEPTS AND EARLY METHODS (1995-2013)

In fact, the theoretical and algorithmic foundations for modern object detection were laid during this period. Key developments included Support-Vector Machines (SVMs) [27], proposed by Cortes and Vapnik, which became a popular classifier in early detection systems. The concept of spatial pyramid matching (SPM) [28], introduced by Lazebnik et al., provided a crucial multi-scale representation for images that was later adopted by detection architectures. A major catalyst for progress was the creation of the large-scale ImageNet dataset [29] by Deng et al., which became

the foundational benchmark for pre-training deep networks. This era also saw key optimization strategies emerge, such as the Rectified Linear Unit (ReLU) activation function [30] and applications of the Sinkhorn-Knopp algorithm for optimal transport [31]. Finally, a pivotal moment was the 2013 introduction of Selective Search by Uijlings et al. [32]. This heuristic region proposal algorithm moved beyond exhaustive search methods and directly enabled the first generation of efficient CNN-based detectors.

B. THE CNN REVOLUTION AND TWO-STAGE DOMINANCE (2014-2015)

This period saw the successful integration of deep CNNs into object detection, establishing the two-stage paradigm. In 2014, Girshick et al. proposed R-CNN [5], which combined Selective Search proposals with deep features extracted by a CNN (like AlexNet) and class-specific SVM classifiers. Concurrently, Simonyan and Zisserman developed VGGNet [33], a very deep CNN that became a standard backbone. Zeiler and Fergus's work on visualizing CNNs [34] also provided critical insights.

To address R-CNN's computational bottleneck, He et al. introduced Spatial Pyramid Pooling Networks (SPPNet) [35] in 2015. SPPNet applied the CNN once per image and used a spatial pyramid pooling layer to extract fixed-length features from arbitrary-sized region proposals, significantly accelerating the pipeline. Building on this, Girshick proposed Fast R-CNN [36] later that year, which integrated region proposal warping with an Region of Interest (RoI) Pooling layer and replaced SVMs with a softmax layer within a single, end-to-end trainable network. Meanwhile, Szegedy et al. introduced GoogLeNet [37], featuring inception modules that would influence future architectures.

C. THE RISE OF ANCHOR-BASED AND REAL-TIME DETECTION (2016)

2016 was a watershed year with the introduction of anchor-based mechanisms and efficient one-stage detectors. Ren et al. proposed Faster R-CNN [6], which replaced the slow Selective Search with a learnable Region Proposal Network (RPN). The RPN introduced *anchor boxes*—predefined bounding boxes of various scales and aspect ratios—as regression references, creating a fully convolutional, end-to-end two-stage detector. Dai et al. followed with Region-based Fully Convolutional Network (R-FCN) [38], a fully convolutional variant that improved speed by generating position-sensitive score maps.

Simultaneously, the one-stage paradigm emerged for real-time applications. Redmon et al. introduced YOLO (You Only Look Once) [7], which framed detection as a single regression problem on an image grid, achieving remarkable speed. Liu et al. proposed SSD (Single Shot MultiBox Detector) [9], which combined YOLO's grid approach with anchor boxes from Faster R-CNN and made predictions from multiple feature maps for better scale handling. This year

also saw the introduction of ResNet [39] by He et al., whose residual connections became a foundational backbone for future detectors.

D. ARCHITECTURAL REFINEMENTS AND ADDRESSING IMBALANCE (2017)

Research focused on enhancing feature representation and tackling the foreground-background class imbalance inherent in one-stage detectors. Lin et al. introduced Feature Pyramid Networks (FPN) [18], which constructed a multi-scale feature pyramid with rich semantics through top-down and lateral connections, greatly improving detection across scales. He et al. extended Faster R-CNN to Mask R-CNN [17] for instance segmentation, replacing RoI Pooling with the more precise RoIAlign.

For one-stage detectors, the critical issue of class imbalance was addressed. Lin et al. proposed RetinaNet [14] and its novel *Focal Loss*, which down-weighted the loss from easy negative samples, allowing the model to focus on hard examples and closing the accuracy gap with two-stage methods. Redmon and Farhadi released YOLOv2/YOLO9000 [40], incorporating anchor boxes, batch normalization, and multi-scale training. Other improvements included DSSD [41] (using deconvolution), DSOD [42] (training from scratch), and A-Fast-RCNN [43] (adversarial training).

E. MULTI-SCALE ENHANCEMENT AND ANCHOR-FREE EXPLORATION (2018)

The field diversified with sophisticated multi-scale architectures and the first explorations of anchor-free detection. Liu et al. proposed PANet [44], adding a bottom-up path augmentation to FPN for better information flow in instance segmentation. Numerous works improved feature fusion and context aggregation, such as HyperNet [45], Parallel FPN [46], and Deep Feature Pyramid Reconfiguration [47]. Refinements in training included Scale Normalization for Image Pyramids (SNIP) [48] for scale normalization and Libra R-CNN [49] for balanced learning.

A major shift began with the introduction of anchor-free detectors. Law and Deng proposed CornerNet [50], detecting objects as pairs of keypoints (top-left and bottom-right corners) using associative embedding, eliminating anchor boxes entirely. For one-stage detectors, refinement strategies like RefineDet [51] and feature enrichment methods [52], [53], [54] were proposed. YOLOv3 [55] was released, featuring a deeper Darknet backbone, multi-scale predictions, and independent logistic classifiers. Other notable 2018 works include MegDet [56] for large-batch training, Guided Anchoring [57], MetaAnchor [58], and Dynamic Anchor Feature Selection [59].

F. CONVERGENCE OF PARADIGMS AND EFFICIENT MODELS (2019)

This year saw anchor-free methods mature and a focus on model efficiency. Multiple anchor-free approaches were

proposed: ExtremeNet [60] detected four extreme points and a center point; RepPoints [61] represented objects as adaptive point sets; FSAF [62] introduced an anchor-free module for feature selection; and FCOS [63] performed per-pixel prediction with a center-ness branch. The two-stage TridentNet [64] used scale-specific branches with dilated convolutions.

Efficiency became a key theme. Tan and Le introduced EfficientNet [65], using compound scaling to achieve better accuracy with fewer parameters. Loss function improvements included AP-Loss [66] and bounding box regression with uncertainty [67]. Training strategies advanced with ScratchDet [68] and AutoFocus [69]. Improved proposal methods like Learning to Rank Proposals [70] were also introduced.

G. NEURAL ARCHITECTURE SEARCH AND ADAPTIVE ASSIGNMENT (2020)

Neural Architecture Search (NAS) and adaptive training strategies came to the forefront. Du et al. proposed SpineNet [71], a backbone with a scale-permuted architecture learned via NAS. Tan et al. built upon EfficientNet with EfficientDet [19], featuring a weighted BiFPN and compound scaling. Kim and Lee introduced Probabilistic Anchor Assignment (PAA) [72], formulating anchor assignment as a probability distribution.

A critical bridge between anchor-based and anchor-free methods was presented by Zhang et al. with Adaptive Training Sample Selection (ATSS) [73], which automatically selected positive/negative samples based on object statistics. The YOLO series continued with the release of YOLOv5 (Ultralytics, 2020), featuring a CSPDarknet backbone and a PANet-like neck. The transformer era also began, with Dosovitskiy proposing the Vision Transformer (ViT) [74] for image classification, and Carion et al. introducing DETR [10], the first end-to-end transformer-based object detector that eliminated anchors and NMS using bipartite matching loss.

H. THE TRANSFORMER ERA AND ADVANCED ASSIGNMENTS (2021)

Transformers rapidly evolved to address DETR's slow convergence and were combined with CNNs. Liu et al. proposed the Swin Transformer [75], a hierarchical vision transformer with shifted windows that became a powerful backbone. To accelerate DETR, Gao et al. introduced Spatially Modulated Co-Attention (SMCA) [76], and Wang et al. proposed Anchor DETR [77], using anchor points to give object queries spatial meaning. Ghiasi et al. demonstrated the effectiveness of a simple Copy-Paste data augmentation strategy [78].

Advanced label assignment strategies continued to develop. Ge et al. formulated label assignment as an Optimal Transport problem with OTA [79], using the Sinkhorn-Knopp

algorithm [31]. The YOLO lineage saw Scaled-YOLOv4 [80], optimizing cross-stage partial networks.

I. SPLIT ATTENTION AND DYNAMIC LABEL ASSIGNMENT (2022)

Refinements in transformer architectures and label assignment mechanisms marked this year. He and Todorovic proposed DESTRA [81], which split the cross-attention in transformers into separate branches for classification and regression. For anchor-free detection, Su et al. introduced Dynamic Smooth Label Assignment (DSLAs) [82], which improved the transition between positive and negative samples using a smooth IoU-based label.

J. ARCHITECTURAL REFORMS AND COMPOUND SCALING (2023)

The trend towards highly optimized, efficient architectures continued. Wang et al. introduced YOLOv7 [83], a real-time detector employing architectural reforms and a “bag-of-freebies” training strategy to achieve state-of-the-art speed and accuracy. The YOLO series further advanced with the release of YOLOv8 (Ultralytics, 2023), which featured a revised architecture with a C2f module and supported multiple vision tasks.

K. HYBRID ARCHITECTURES AND THE MATURING YOLO LINEAGE (2024)

The years 2024 and 2025 witnessed the rapid, concurrent evolution of the YOLO family, solidifying its architectural dominance for real-time detection while introducing significant theoretical and engineering improvements. In early 2024, Wang et al. introduced YOLOv9 [84], which addressed the fundamental issue of information loss in deep networks through *Programmable Gradient Information (PGI)*. PGI provided complete input information to the target task’s objective function, enabling more reliable gradient flow for weight updates and allowing models trained from scratch to achieve state-of-the-art performance. Coupled with a new *Generalized Efficient Layer Aggregation Network (GELAN)* backbone, YOLOv9 enhanced parameter utilization without relying on depth-wise convolutions. Shortly after, YOLOv10 [85] was released, focusing on post-processing efficiency by introducing *consistent dual assignments* for both one-to-one and one-to-many matching during training. This innovation enabled *Non-Maximum Suppression (NMS)-free inference*, eliminating a persistent computational bottleneck and further boosting end-to-end speed.

The lineage advanced with YOLOv11 [86], unveiled in late 2024, which introduced key architectural refinements for enhanced feature extraction and efficiency. Its backbone and neck replaced the previous C2f block with the more efficient *C3k2 block*, utilizing two smaller convolutions for faster processing. A significant addition was the *C2PSA (Convolutional block with Parallel Spatial Attention)* module, which incorporated a spatial attention mechanism to help the model

focus on more informative regions within the feature maps. This period also saw the YOLO framework’s capabilities expand beyond bounding box detection to become a versatile platform supporting instance segmentation, pose estimation, and oriented object detection (OBB). Comparative analyses in applied domains like smoke and fire detection demonstrated the continuous performance gains of these iterations, with YOLOv11 variants often achieving superior precision and mean Average Precision (mAP) metrics.

L. THE REAL-TIME TRANSFORMER CHALLENGE AND OPEN-SET DETECTION (2024)

While YOLO variants progressed, transformer-based architectures made a decisive entry into the high-speed detection arena. Zhao et al. proposed *RT-DETR* [21], the first real-time end-to-end object detector built on the DETR framework. RT-DETR tackled the core speed limitations of prior DETRs through an efficient hybrid encoder that decoupled intra-scale interaction and cross-scale fusion. It also introduced an *uncertainty-minimal query selection* scheme to provide higher-quality initial queries to the decoder. Critically, by its end-to-end nature, RT-DETR completely eliminated the need for NMS. The model demonstrated that DETRs could surpass comparable YOLO models in both speed and accuracy on the COCO benchmark, marking a pivotal moment where transformer detectors became competitive for real-time applications.

Concurrently, the field saw significant strides toward more generalizable, open-world perception. The trend of marrying DETR-like architectures with large-scale vision-language pre-training culminated in models like *Grounding DINO* [87]. By fusing a transformer-based detector with grounded pre-training, it achieved open-set object detection, capable of detecting arbitrary objects based on human inputs such as category names or referring expressions. This line of work advanced with *Grounding DINO 1.5* in 2024, which introduced a *Pro* model for stronger generalization and an *Edge* model optimized for deployment. Trained on over 20 million images with grounding annotations, these models set new records on zero-shot detection benchmarks, pushing the “edge” of what was detectable without task-specific fine-tuning.

M. ADVANCED INTERACTION AND RELATIONSHIP DETECTION (2024-2025)

The contemporary research landscape has evolved from isolated object localization toward the comprehension of complex semantic relationships between entities. Human-Object Interaction (HOI) detection [88], which predicts $\langle \text{human}, \text{action}, \text{object} \rangle$ triplets, represents a crucial frontier that requires more advanced visual understanding than traditional detection.

A primary challenge identified in recent literature is the phenomenon of **interaction information loss**. As noted by the RmLR framework [89], traditional two-stage detectors

often generate entity tokens that prioritize spatial coordinates and object categories while failing to capture crucial relational cues. To mitigate this, RmLR proposes a **re-mining** strategy via an Interactive Relation Encoder (IRE) to extract interaction-aware features directly from low-level visual tensors. This is complemented by a fine-grained word-level alignment strategy that manages many-to-many matching between multiple concurrent interactions and their textual descriptions.

Simultaneously, the pursuit of **universal** detectors has led to the development of UniHOI, which addresses the **lack of open-world knowledge** inherent in models constrained by discrete labels. UniHOI leverages Vision-Language (VL) foundation models (e.g., BLIP-2 [90]) and Large Language Models (LLMs) [91] to support open-category recognition. By utilizing a Human-Object (HO) Prompt-guided Decoder, UniHOI effectively queries high-level relation representations from foundation models, enabling the system to understand intricate behaviors through flexible textual knowledge retrieval.

1) POSITIONING THE UOD-DSM CONTRIBUTION

The proposed UOD-DSM addresses a critical **universal optimality** gap that remains even within these advanced semantic paradigms. While RmLR and UniHOI optimize for relational rigor and cross-modal alignment, they are often characterized by high computational demands or fixed architectural paths. The UOD-DSM provides the necessary meta-architectural layer to operationalize these findings. By utilizing the DSM routing layer, the system can dynamically activate a Transformer-based branch—favored for the global attention required in interaction reasoning—when high scene complexity is detected, or revert to efficiency-optimized one-stage pathways for less demanding real-time contexts. This approach transforms the static architectural commitment of current detectors into an adaptive, paradigm-aware framework.

N. TOWARD FOUNDATIONAL MODELS AND EDGE EFFICIENCY (2024-2025)

The period from 2024 into 2025 has been characterized by the convergence of several trends: the pursuit of foundational vision models, the relentless drive for edge device efficiency, and the blurring of boundaries between core vision tasks. While not a detector per se, the introduction of the *Segment Anything Model (SAM)* [92] in 2023 created a profound impact. Its promptable design and massive dataset (SA-1B) established a new paradigm for foundational vision models, inspiring numerous subsequent approaches that adapted or leveraged its powerful segmentation capabilities for detection-related tasks.

In practice, the dominant theme has been *specificity through efficiency*. Research and development have increasingly focused on creating not just universally accurate models, but families of models (like YOLOv11's nano to

extra-large variants or Grounding DINO 1.5's Pro/Edge split) tailored for specific deployment scenarios. The optimization for edge computing, mobile platforms, and specialized industrial applications (e.g., aerial imagery, medical imaging, embedded systems) has become a primary driver of architectural innovation. Furthermore, the leading detection frameworks have evolved into multi-task learning platforms, efficiently handling object detection, instance segmentation, and pose estimation within a unified architecture, signaling a move towards more capable and efficient general-purpose vision systems.

O. COMPARATIVE ANALYSIS OF SPECIALIZED AND ADAPTIVE ARCHITECTURES

The evolution of deep learning has shifted from static, single-purpose models toward systems characterized by active adaptation, sequential intelligence, and structured feature learning. This research is positioned at the intersection of these trends, seeking to overcome the “universal optimality gap” where specialized architectures excel in isolated domains but fail to maintain performance across heterogeneous application contexts.

1) COMPARISON WITH SPECIALIZED DEEP LEARNING APPROACHES

Recent scholarship has introduced various mechanisms for improving model performance through targeted learning strategies. For instance, the **Deep Learning Based Active Learning (DLBAL)** method [93] utilizes an EfficientNet-B0 backbone to minimize human intervention by automatically selecting high-confidence samples for labeling, thereby optimizing the training process for image classification. Similarly, research into **Violent Scene Detection in Cartoon Movies** [94] employs a combined CNN-LSTM architecture to capture the spatiotemporal dynamics necessary for accurate video classification, utilizing shot segmentation to manage computational costs. Furthermore, the **Quality-Centric Embedding and Ranking Network (QCERN)** [95] for Blind Image Quality Assessment (BIQA) utilizes Adaptive Ranking Transformers (ART) and score pivots to cluster images in a structured embedding space based on perceptual quality.

While these works provide critical advancements in adaptation—specifically through dataset refinement, temporal sequence modeling, and quality-aware feature ranking—they remain fundamentally constrained by their reliance on a fixed architectural paradigm. The DLBAL system optimizes the *data distribution* for a static classifier, and the cartoon detection model provides a specialized solution for *sequential temporal data*. In contrast, the UOD-DSM introduces a meta-architectural layer that dynamically switches between fundamentally different detection families (one-stage, two-stage, and Transformers) at runtime. Unlike the quality-centric focus of QCERN, which organizes features for

assessment, the UOD-DSM organizes detection paradigms themselves for operational efficiency and precision.

2) PAPER MOTIVATION

The primary motivation for this work arises from the observation that specialized models, while effective within their respective domains (such as sequence-based violence detection or quality-centric embedding), cannot bridge the performance gap when operational requirements fluctuate between high-speed surveillance and precision-critical medical imaging. Current solutions are either static or only adaptively efficient within a narrow architectural family, failing to leverage the full spectrum of strengths offered by the diverse historical evolution of object detection. This paper is motivated by the need for a “fully adaptive vision system” that can interpret metadata and scene complexity to automatically align its internal strategy with the immediate operational context.

3) NOVELTY AND ADVANTAGES

The novelty of the UOD-DSM architecture over existing specialized works lies in its two complementary contributions:

- 1) **Data-Driven Bibliometric Mapping:** Unlike models guided by heuristic judgment, the UOD-DSM’s routing logic is derived from a systematic analysis of 55,868 documents to establish an empirical “Paradigm–Domain Relationship Matrix”.
- 2) **Meta-Architectural Adaptivity:** While prior works like the CNN-LSTM sequential model or the ART-based ranking network are “paradigm-static,” the UOD-DSM is “paradigm-aware”. It provides a **DSM** that functions as a knowledge-driven routing layer to activate the optimal sub-network—such as a Transformer branch for global attention or a one-stage pathway for real-time throughput—based on real-time triggers.

By integrating these disparate branches within a unified framework, the UOD-DSM offers a unique advantage: it manages the computational overhead of large-scale, high-precision models by reverting to efficiency-optimized pathways for less demanding contexts, thereby achieving a “universal optimality” that single-paradigm systems cannot replicate.

III. METHODOLOGY

A. AIM 1

This section outlines the methodology designed to achieve Aim 1, which seeks to identify the most influential object detection paradigms and their application domains using a data-driven bibliometric approach. This method avoids reliance on predefined categories, allowing natural patterns and research clusters to emerge directly from the bibliographic data. The analysis integrates quantitative citation indicators and co-occurrence mapping using VOSviewer.

TABLE 1. Scopus search query used for dataset retrieval (2014–2025).

Component Type	Query Segment
Key Detection Terms	TITLE-ABS-KEY ("object detection" OR "object detector" OR "object recognition")
Key Paradigm Terms	AND TITLE-ABS-KEY ("deep learning" OR "cnn" OR "transformer" OR "vision" OR "detection framework")
Document Type Limit	AND (LIMIT-TO (DOCTYPE , "ar") OR LIMIT-TO (DOCTYPE , "cp"))
Subject Area Limit	AND (LIMIT-TO (SUBJAREA , "COMP") OR LIMIT-TO (SUBJAREA , "ENGI") OR LIMIT-TO (SUBJAREA , "MATH") OR LIMIT-TO (SUBJAREA , "PHYS") OR LIMIT-TO (SUBJAREA , "DECI"))
Publication Year Limit	AND (LIMIT-TO (PUBYEAR , 2014) OR LIMIT-TO (PUBYEAR , 2015) OR LIMIT-TO (PUBYEAR , 2016) OR LIMIT-TO (PUBYEAR , 2017) OR LIMIT-TO (PUBYEAR , 2018) OR LIMIT-TO (PUBYEAR , 2019) OR LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2021) OR LIMIT-TO (PUBYEAR , 2022) OR LIMIT-TO (PUBYEAR , 2023) OR LIMIT-TO (PUBYEAR , 2024) OR LIMIT-TO (PUBYEAR , 2025))
Language Limit	AND (LIMIT-TO (LANGUAGE , "English"))

1) DATA COLLECTION

Bibliographic data were retrieved from the Scopus database for the period 2014–2025. A general query was employed to capture all deep learning–based object detection research, ensuring the resulting dataset encompasses various architectural paradigms and domains without any pre-assigned filters. The specific complex query structure utilized to achieve this broad, yet focused, retrieval is detailed in table 1.

The retrieved records were exported on **22-10-2025** in CSV format, including bibliographic metadata such as titles, abstracts, author keywords, citations, and references. The exported data comprised detailed fields covering four key areas: 1/ Citation information (Author(s), Document title, Year, EID, Source title, Volume, issues, pages, Citation count, Source & document type, Publication stage, DOI, Open access); 2/ Bibliographical information (Language of original document); 3/ Abstract & keywords (Abstract, Author keywords, Indexed keywords); and 4/ Other information (Include references). This comprehensive dataset serves as the foundation for subsequent clustering, visualization, and trend identification. This dataset serves as the foundation for the subsequent clustering, visualization, and trend identification.

2) DATA HANDLING AND SEGMENTATION

Due to Scopus export limitations inherent in handling large dataset volumes, the total corpus of 55,868 documents was divided into four distinct subsets to ensure full coverage

across the twelve-year span. This segmentation allowed for efficient handling and subsequent bibliometric mapping using VOSviewer, enabling reliable co-occurrence, citation, and clustering analyses across time. Each subset preserved the same filtering criteria, ensuring methodological consistency. The data was subsequently merged to ensure methodological consistency and full coverage across the twelve-year span.

The dataset was segmented as follows:

- 1) **2014–2015:** 1,759 documents
- 2) **2016–2021:** 19,030 documents
- 3) **2022–2023:** 16,527 documents
- 4) **2024–2025:** 18,552 documents

3) EXPLORATORY MAPPING USING VOSVIEWER

The mapping procedure identified research themes and relationships among keywords. A co-occurrence network of author keywords was created using full counting, with the minimum keyword occurrence threshold set to 5. Applying this threshold to the initial 65,500 unique author keywords resulted in 4,528 valid keywords. To ensure analytical depth and readability, the 1,000 most connected keywords (those with the highest Total Link Strength (TLS)) were selected for clustering and visualization. Each resulting cluster represents a distinct thematic area, potentially corresponding to an object detection paradigm or application domain.

Clusters were interpreted manually based on their most frequent and connected keywords, allowing for the identification of paradigms and domains directly from the bibliometric data. For example, a cluster containing YOLO, SSD, real-time, vehicle represents one-stage or real-time detection, while a cluster with R-CNN, segmentation, Computed Tomography (CT), tumor represents two-stage detectors in medical imaging. Overlay visualization utilized citation averages as the overlay weight, where yellow or red nodes indicate high-impact keywords (highly cited), highlighting the most influential paradigms and domains.

4) CITATION IMPACT AND LINK STRENGTH EVALUATION

Quantitative indicators were extracted for each cluster to assess influence and connectivity. Both TLS, which measures influence and connectivity, and Average Citations per Keyword, which measures scholarly impact, were employed.

5) IDENTIFICATION OF PARADIGM–DOMAIN CLUSTERS

The top clusters identified by VOSviewer correspond to emergent relationships between object detection paradigms and their application domains, discovered directly from data. Examples include Cluster 3 (DETR, transformer, Uncrewed Aerial Vehicle (UAV), satellite) representing Transformer / aerial sensing. These emergent clusters serve as the empirical foundation for the Results and Discussion (Aim 1) section.

6) METHODOLOGICAL RATIONALE

This data-driven approach eliminates the bias associated with predefined classifications. The combined use of

co-occurrence mapping and citation overlay reliably identifies both popular paradigms (by frequency) and influential paradigms (by citation impact). This provides a reliable empirical base for determining domain-specific strengths and supporting the conceptual development needed for Aim 2.

B. AIM 2

The architecture proposed for Aim 2 represents the Conceptual Design of the UOD utilizing a DSM. This design systematically builds upon the empirical bibliometric findings of Aim 1, which rigorously established that the optimal object detection paradigm is domain-contingent; that is, peak performance is achieved when a specific detection paradigm (one-stage, two-stage, or transformer-based) is utilized for its corresponding domain. Accordingly, the proposed methodology integrates these specialized paradigms into a single unified framework, designed to dynamically adapt to specific domain requirements. The core methodological goal of Aim 2 is the operationalization of the paradigm–domain relationships discovered in Aim 1 into a functional, adaptable detection system.

1) OVERVIEW AND OBJECTIVE

The second methodological objective (Aim 2) focuses on the development of a UOD that integrates the strengths of multiple detection paradigms—one-stage, two-stage, and transformer-based—within a single adaptable architecture. This unified design is guided by the empirical findings of Aim 1, which demonstrated that the optimal detection paradigm depends on task-specific priorities, such as speed, precision, or contextual understanding. The proposed UOD framework employs a DSM that automatically selects and activates the most suitable paradigm at runtime based on the characteristics of the input domain or application requirements.

2) CONCEPTUAL DESIGN OF THE UNIFIED DETECTOR

The conceptual architecture of the UOD systematically operationalizes the fundamental principle that distinct application domains necessitate specialized detection paradigms. Figure 1 illustrates the conceptual block architecture of the UOD, featuring the DSM as its decision-making core. The framework is organized into four sequential and functionally integrated stages that operationalize a domain-adaptive detection workflow.

This structured design ensures that internal routing and paradigm selection remain explicitly aligned with the empirically derived domain requirements established through the preceding bibliometric analysis. The architecture formally transitions from high-level requirement interpretation (Stage I), through knowledge-guided dynamic detector selection (Stage II) and trigger-based paradigm activation (Stage III), to the generation of standardized detection outputs (Stage IV).

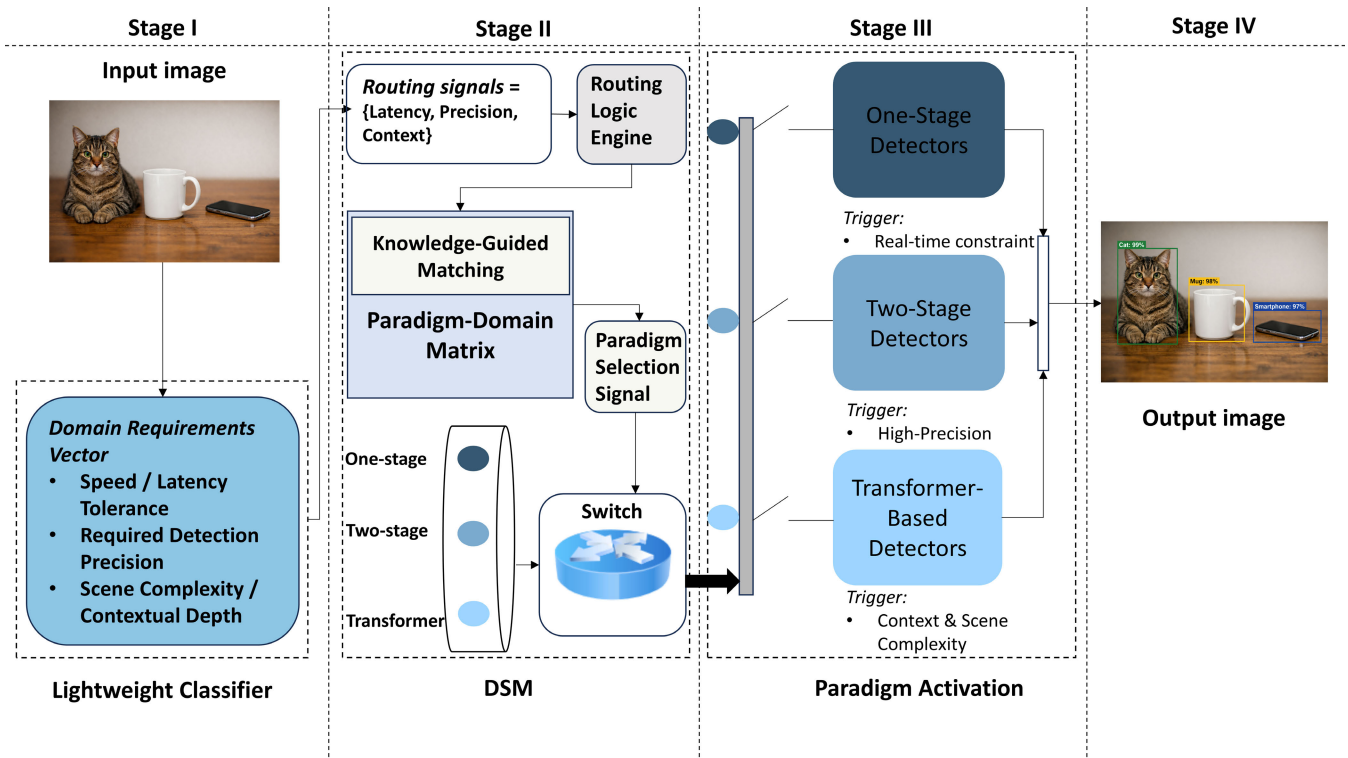


FIGURE 1. Conceptual block architecture of the UOD utilizing a DSM. The figure illustrates the sequential flow of operations across four stages, culminating in domain-adaptive detection results. Example input and output images shown are AI-generated and are included solely for illustrative and visualization purposes.

a: STAGE I – INPUT AND REQUIREMENT INTERPRETATION

The process commences when an input image or sequence is introduced into the system. This initial stage is managed by a Lightweight Classification Module that performs requirement interpretation. This module analyzes contextual metadata or extracted visual features to infer application-specific operational priorities. These critical inferred parameters include the evaluation of speed and latency tolerance (characteristic of real-time systems), the required detection precision (critical for medical imaging), and scene complexity or contextual depth (relevant to aerial imagery). This analysis facilitates the mapping of these attributes to the corresponding application domain (e.g., autonomous systems, remote sensing), operationalizing the domain inference logic necessary for subsequent paradigm selection.

b: STAGE II – DYNAMIC DETECTOR SELECTION (THE DSM LAYER)

The Dynamic Switch Mechanism (DSM) constitutes the decision-making core of the unified system, functioning as a knowledge-driven routing layer that determines which object detection paradigm should be activated. The DSM references the Paradigm–Domain Matrix established in Aim 1, which empirically encodes the optimal detector paradigms relative to specific domain requirements based on bibliometric co-occurrence and citation strength analysis. Functionally, the

switch utilizes the domain features identified in Stage I to select the most appropriate detection paradigm.

The DSM layer performs three essential functions: (1) Domain Matching, cross-referencing interpreted input requirements against the empirical matrix; and (2) Decision Routing, selecting the appropriate paradigm branch (one-stage, two-stage, or transformer-based).

c: STAGE III – PARADIGM ACTIVATION LOGIC

Upon selection by the DSM, the Paradigm Activation Layer triggers the corresponding specialized detector sub-module. The activation logic is dictated by the empirical findings of Aim 1. This detailed selection criteria is summarized in table 2. Based on the table:

- One-Stage Detectors are used when facing a high-speed, real-time constraint; their objective is to minimize latency and maximize throughput, typically associated with autonomous and surveillance systems.
- Two-Stage Detectors are triggered by the need for high-precision and fine localization. Their primary goal is to maximize accuracy and recall in domains like medical and inspection applications.
- Transformer-Based Detectors are selected when there is high context and scene complexity, with the primary objective to model long-range dependencies, often applied in remote sensing and UAV imagery.

TABLE 2. Paradigm activation logic within the unified detector, dictated by Aim 1 findings.

Paradigm Type	Activation Trigger	Primary Objective	Typical Domain Association
One-Stage Detectors	High-speed, real-time constraint	Minimize latency and maximize throughput	Autonomous and surveillance systems
Two-Stage Detectors	High-precision, fine localization	Maximize accuracy and recall	Medical and inspection applications
Transformer-Based Detectors	High context and scene complexity	Model long-range dependencies	Remote sensing and UAV imagery

Each specialized sub-detector remains independently optimized for its specific objective but operates cohesively under the unified framework through the shared dynamic switch mechanism, ensuring interoperability and consistency.

d: STAGE IV – OUTPUT GENERATION

Following paradigm activation, the selected detector processes the input image and produces the final detection results, including bounding boxes, class labels, and confidence scores. A Result Synthesis Module then consolidates these detections into a standardized format. This unified output ensures consistency across different detection paradigms, simplifies post-analysis, and maintains compatibility with downstream tasks such as tracking or segmentation. The complete workflow is illustrated conceptually in Figure 1.

3) METHODOLOGICAL RATIONALE

This hybrid design systematically bridges the empirical bibliometric insights derived from Aim 1 with computational implementation in Aim 2. This methodology ensures: objectivity, as paradigm–domain associations are data-driven and not arbitrarily assigned; adaptivity, as detector selection is dynamically informed by domain characteristics; and scalability, allowing for the integration of future paradigms (e.g., diffusion-based or graph-based detectors). By translating bibliometric mappings into algorithmic decision rules, this approach introduces a novel paradigm-aware framework that adapts to the contextual demands of any application domain.

IV. RESULTS AND DISCUSSIONS

A. RESULTS FOR AIM 1

1) DATASET OVERVIEW AND FILTERING

The bibliometric search retrieved a total of **55,868 documents** published between 2014 and 2025, obtained from Scopus using the defined query. The initial corpus covered 24 subject areas. To focus on the core scientific and methodological aspects, only the five dominant subject areas were retained for analysis:

- **Computer Science:** 47,269 documents.
- **Engineering:** 29,836 documents.
- **Mathematics:** 14,624 documents.
- **Physics and Astronomy:** 8,738 documents.
- **Decision Sciences:** 5,324 documents.

Note that the sum of the subject counts (105,791) is greater than the total number of documents (55,868)

because one document can belong to multiple subject areas. These five disciplines collectively represent 83.3% of the total corpus. Consequently, the remaining 20 subject areas were systematically excluded to maintain analytical consistency and prevent topic dilution beyond the computational and algorithmic core of the research.

Figure 2 present, detailing the proportional distribution of documents per subject area.

2) TEMPORAL DISTRIBUTION OF PUBLICATIONS (2014–2025)

Figure 3 illustrates the temporal distribution of annual publications, indicating sustained expansion of the research field between 2014 and 2024, increasing from 729 publications in 2014 to a peak of 10,404 in 2024. The early formative phase (2014–2018) shows steady growth, rising from 729 to 2,379 publications. A marked acceleration begins in 2019 (3,792 publications), followed by continued expansion through 2022 (7,392) and 2023 (9,135), culminating in the highest annual output in 2024. Although 2025 records 8,148 publications, the overall trajectory across the twelve-year period confirms the sustained structural maturation of the domain.

To further characterize research momentum, the Annual Growth Rate is computed as:

$$\text{Growth Rate}_t = \frac{N_t - N_{t-1}}{N_{t-1}} \times 100 \quad (1)$$

where N_t denotes the number of publications in year t .

The growth-rate curve, also presented in Figure 3, highlights distinct dynamics, including peak expansion in 2017 (51.88%) and 2019 (59.39%), followed by moderated but sustained acceleration between 2020 and 2022 (15.14%–33.17%). The negative growth rate observed in 2025 (−21.68%) should be interpreted cautiously, as the bibliographic records were retrieved on 22 October 2025. Consequently, the 2025 data reflect partial-year indexing rather than a confirmed annual contraction.

The inclusion of exact numerical values for each data point enhances quantitative transparency and enables precise year-by-year comparison of both absolute output and relative growth behavior.

3) THRESHOLDING AND NETWORK PREPARATION

Following preprocessing and normalization, the dataset initially contained 65,500 unique author keywords. Applying the minimum keyword occurrence threshold of five resulted

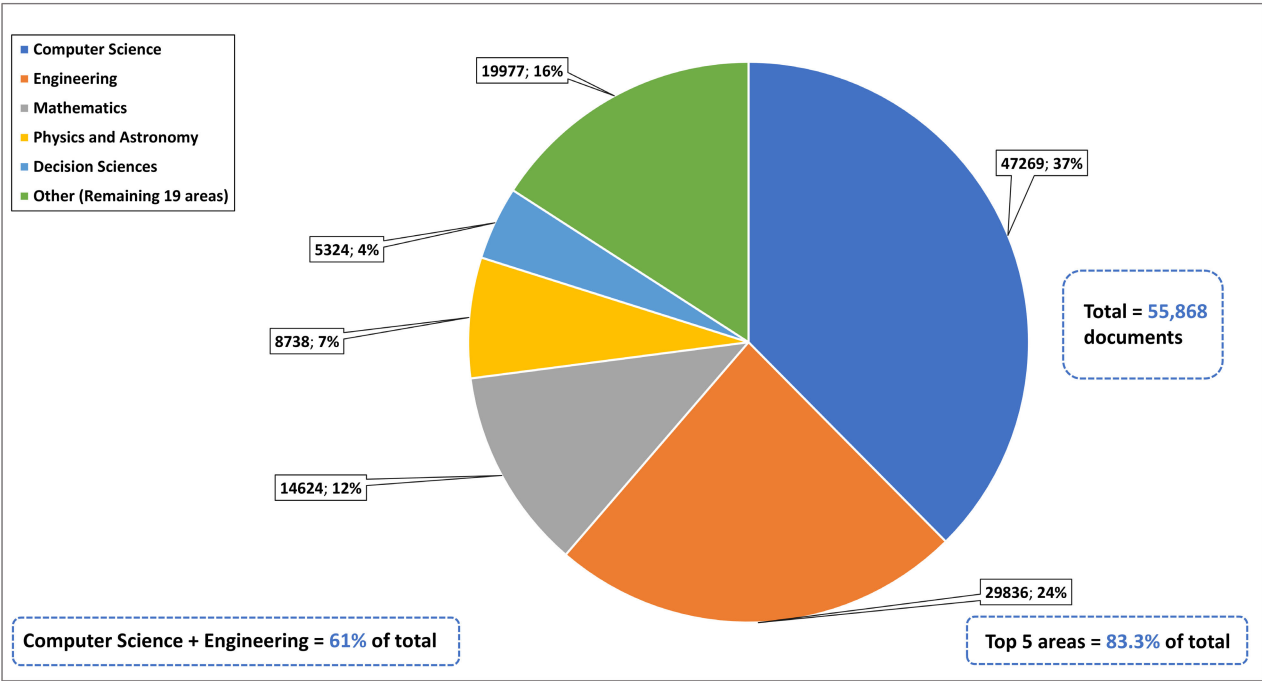


FIGURE 2. Proportional distribution of documents per subject area.

in 4,528 valid keywords. VOSviewer computed the TLS for each keyword. To ensure analytical depth and readability, the 1,000 most connected keywords (those with the highest TLS) were selected for clustering and visualization.

4) NETWORK AND CLUSTER FORMATION

The resulting co-occurrence network identified 14 distinct clusters, with nine major clusters demonstrating substantial internal connectivity and thematic coherence. Clusters 1, 3, and 4 exhibited the highest TLS values, suggesting their central role in the research network structure.

Table 3 present, summarizing the Major Cluster Metrics from VOSviewer Analysis. These metrics show that Cluster 1 (Core Detection & Learning Paradigms) had 58 keywords and the highest TLS at **15,891**. Cluster 3 (Detection Architectures & Applications, TLS 11,468) and Cluster 4 (Remote Sensing & Complex Environments, TLS 10,073) also demonstrated high TLS values.

5) HIGH-IMPACT KEYWORDS AND CENTRAL TOPICS

The field’s conceptual backbone is defined by the most frequent and connected keywords. **Object detection** (16,071 occurrences, 37,017 TLS) and **deep learning** (14,265 occurrences, 33,502 TLS) dominate in both occurrence and link strength, serving as unifying anchors for the network. The prominence of major paradigms like YOLO (1,182 occurrences) and Transformer (1,005 occurrences) confirms the network’s accuracy in reflecting the field’s evolution.

TABLE 3. Major cluster metrics from VOSviewer analysis.

Cluster ID	Theme (Data-Driven Label)	Keywords	Total Occurrences	TLS
1	Core Detection & Learning Paradigms	58	4,967	15,891
2	Tracking & Autonomous Vision	50	3,381	9,132
3	Detection Architectures & Applications	47	3,087	11,468
4	Remote Sensing & Complex Environments	43	3,290	10,073
5	Robotics & Navigation Systems	39	1,969	6,397
6	Human Interaction & Security AI	49	3,634	7,416
7	Edge Computing & Smart Systems	31	1,435	3,600
8	Neural Networks & Dataset Foundations	27	1,459	4,390
9	Advanced Vision Algorithms	13	829	3,617

6) CITATION OVERLAY AND PARADIGM TEMPORAL TRENDS

Overlay visualization using average citation counts showed that clusters linked to **transformer-based paradigms (Clusters 1 and 4) exhibit higher citation impact**, evidenced by yellow nodes in the VOSviewer map. This high citation impact reflects scholarly attention and novelty. Conversely, clusters associated with older, foundational work, such as convolutional and tracking-based paradigms (Clusters 2 and 5), appeared in cooler colors, classifying them as mature and widely deployed technologies.

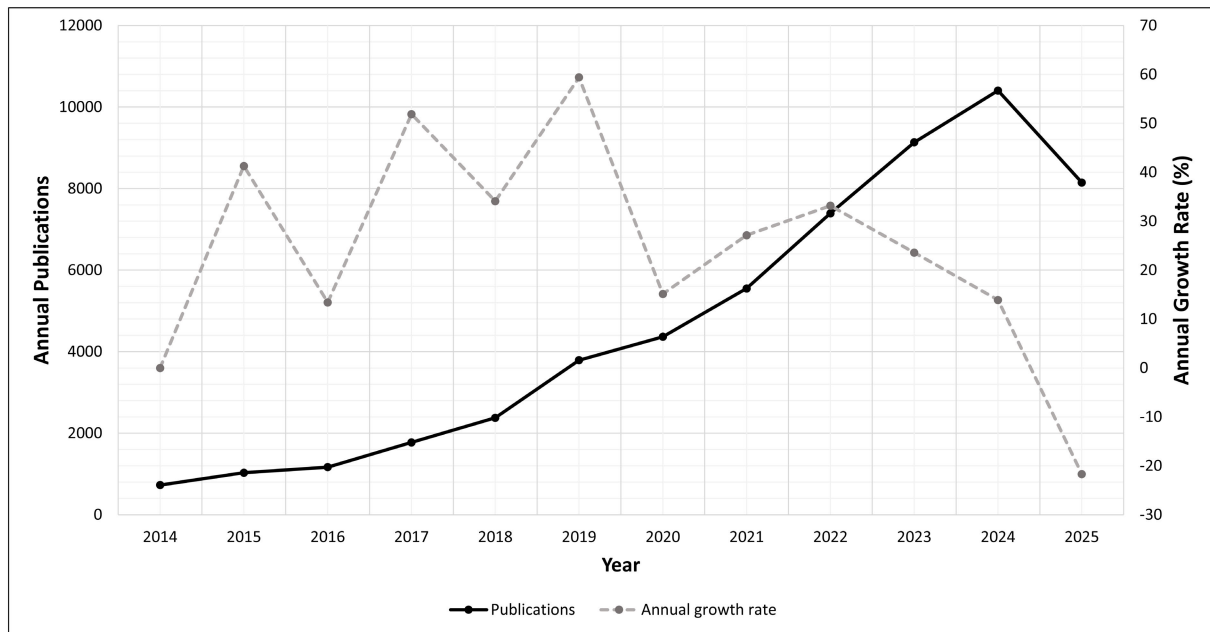


FIGURE 3. Temporal distribution of publications (2014–2025).

Temporal overlay analysis confirmed a clear paradigm shift across the study period:

- **2014–2018:** Dominance of CNN-based and two-stage models.
- **2019–2021:** Emergence of one-stage, real-time architectures.
- **2022–2025:** Rapid rise of transformer-based and hybrid models.

7) PARADIGM–DOMAIN MATRIX

The analysis led to the generation of a Paradigm–Domain Matrix by cross-referencing co-occurrence data with domain-specific keywords. This specialized finding confirms that **the optimal detector paradigm is domain-dependent**, as no single paradigm dominates all domains; suitability hinges on requirements like speed, accuracy, and computational efficiency.

Table 4 present, providing the Paradigm–Domain Matrix Derived from Co-occurrence Analysis. This matrix details specialized findings, such as:

- **One-stage detectors** (e.g., YOLO family) dominate Autonomous / Real-Time Systems, evidenced by strong TLS with “vehicle,” “tracking,” and “real-time” (Clusters 2, 3).
- **Two-stage detectors** (e.g., R-CNN family) excel in Medical Imaging due to association with high-precision, small-object tasks (Clusters 3, 4).
- **Transformer-based architectures** (e.g., DETR, Swin) lead in Remote Sensing / UAV applications, showing strong recent citations (Clusters 1, 4).

8) NETWORK VISUALIZATION AND SUMMARY

The co-occurrence network visualization utilized VOSviewer to map relationships among keywords. Figure 4 present, depicting the VOSviewer Co-occurrence Network of Author Keywords (2014–2025). The map reveals a structured network where nodes are colored by cluster assignment and node size correlates with keyword occurrence. This mapping confirms that object detection and deep learning form the conceptual and citation core of the field.

B. RESULTS FOR AIM 2

This section interprets the conceptual results of the UOD described in Aim 2 and evaluates its theoretical validity in relation to the empirical findings of Aim 1. The discussion emphasizes the functional coherence of the design, its alignment with the data-driven paradigm–domain relationships, and expected performance behaviors under different operational contexts.

1) MAPPING EMPIRICAL FINDINGS TO SYSTEM DESIGN

The table 5 serves as the logical foundation for the DSM. This mapping systematically operationalizes the paradigm–domain relationships derived from bibliometric evidence into algorithmic decision logic within the unified detector.

The mapping demonstrates that the paradigm–domain relationship evidenced by each bibliometric cluster possesses a direct computational counterpart within the unified framework. For instance, when dealing with the

TABLE 4. Paradigm–Domain matrix derived from co-occurrence analysis.

Domain	Dominant Paradigm	Evidence (High TLS & Citation Impact)	Representative Clusters
Autonomous / Real-Time Systems	One-stage (YOLO family)	Strong TLS with “vehicle,” “tracking,” and “real-time”	2, 3
Medical Imaging	Two-stage (R-CNN family)	High citation averages; associated with precision-oriented tasks	3, 4
Remote Sensing / UAV	Transformer-based (DETR, Swin)	Strong recent citations; high rate of late-period publications	1, 4
Industrial / Edge Applications	Hybrid / Fusion Models	High TLS overlap with robotics and embedded systems	5, 7
Security / Surveillance	One-stage detectors	Frequent co-occurrence with “face recognition,” “IoT,” and “tracking”	6

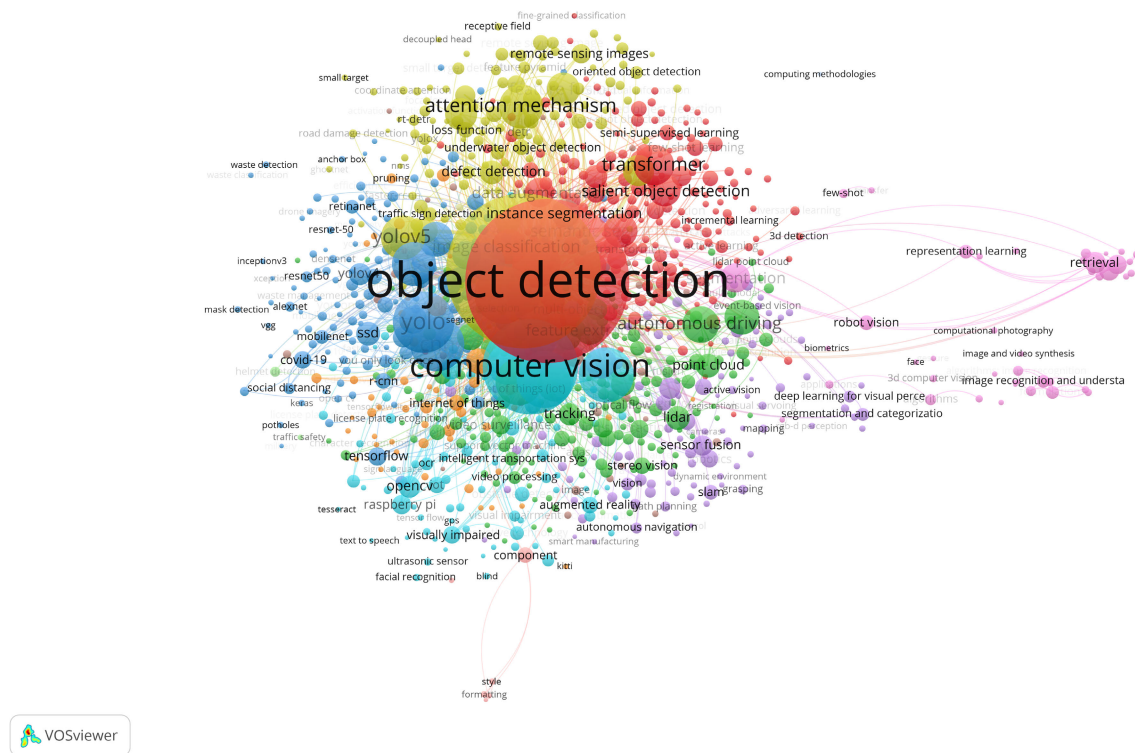


FIGURE 4. VOSviewer co-occurrence network of author keywords (2014–2025). The map reveals a structured network of keywords representing distinct paradigms and application domains in object detection research. Nodes are colored by cluster assignment, and node size correlates with keyword occurrence.

Autonomous / Real-Time Systems empirical domain, the detected dominant paradigm is One-stage (YOLO family), which operationally results in the DSM activating the fast pipeline for the expected behavior of Lowest latency, moderate precision. Conversely, the Medical Imaging domain is associated with the Two-stage (R-CNN family) paradigm, prompting the DSM to activate precision-mode to achieve the expected behavior of Highest accuracy, slower response. Furthermore, domains like Remote Sensing / UAV utilize Transformer-based (DETR, Swin) models, leading the DSM to activate the contextual mode for Best global attention and semantic coverage.

2) EXPECTED SYSTEM PERFORMANCE

The theoretical performance characteristics of the UOD are directly inferred from its architectural design and the rationale governing the DSM. The system is anticipated to yield

adaptive optimization across diverse application domains, conceptually surpassing static architectures. Specifically, adaptivity is governed by the DSM Layer (Stage II), where paradigm selection changes according to the input context. Accuracy is maximized through the activation of two-stage and transformer branches for fine-grained tasks. Speed is maintained by routing high-speed requirements through the low-latency One-Stage Pathway. Furthermore, the global design allows for scalability, permitting additional detectors to be registered in the DSM matrix without necessitating the retraining of the core logic.

3) ILLUSTRATIVE SCENARIO ANALYSIS

Conceptual analysis confirms the coherence between the data-driven analysis and the architectural logic. For instance, in an **Autonomous Navigation** scenario, the DSM triggers the one-stage branch, facilitating millisecond-level inference.

TABLE 5. Operational mapping of empirical paradigm–domain relationships to UOD design.

Empirical Domain (Aim 1)	Dominant Paradigm Detected	Operational Mapping in Aim 2	Expected Behavior
Autonomous / Real-Time Systems	One-stage (YOLO family)	DSM activates fast pipeline	Lowest latency, moderate precision
Medical Imaging	Two-stage (R-CNN family)	DSM activates precision-mode	Highest accuracy, slower response
Remote Sensing / UAV	Transformer-based (DETR, Swin)	DSM activates contextual mode	Best global attention and semantic coverage
Industrial / Edge Applications	Hybrid or Fusion Models	DSM blends lightweight and contextual detectors	Balanced trade-off of speed and accuracy
Security / Surveillance	One-stage lightweight detectors	DSM activates streamlined mode	High throughput with real-time tracking

Conversely, for **Tumor Localization in CT Scans**, the DSM triggers the two-stage branch, enhancing localization accuracy on small lesions, aligning with precision-oriented needs. Finally, in **Satellite Object Detection**, the DSM triggers the transformer branch, enabling the capture of high-altitude contextual dependencies across large images.

C. DISCUSSIONS FOR AIM 1

1) INTERDISCIPLINARY EXPANSION AND NETWORK STRUCTURE

The network analysis (Figure 4) confirms that object detection research functions as a robust, multidisciplinary ecosystem, integrating insights across computer science, engineering, mathematics, physics, and decision sciences. The field's conceptual backbone is anchored by highly central keywords, namely “object detection” (16,071 occurrences, 37,017 TLS) and “deep learning” (14,265 occurrences, 33,502 TLS). **These high TLS values are not merely descriptive; they analytically measure the profound influence and connectivity of these terms within the scholarly network.** The strong internal connectivity within major thematic areas (Cluster 1: 15,891 TLS; Cluster 3: 11,468 TLS; Cluster 4: 10,073 TLS) and the documented cross-links between clusters (e.g., 3–4 and 2–5) reveal significant methodological transfer. This convergence demonstrates that foundational algorithmic development must adapt to domain-specific requirements, such as those found in robotics, healthcare, and remote sensing.

2) PARADIGM–DOMAIN SPECIALIZATION: INTERPRETATION AND ANSWER TO AIM 1

Aim 1 sought to determine the optimal object detector for a specific domain application. The results, synthesized into the Paradigm–Domain Matrix (Table 3), provide the definitive answer: **optimal detector paradigm selection is domain-contingent, confirming that no single architecture universally dominates.**

- 1) **Analysis of One-Stage Dominance:** One-stage detectors (e.g., YOLO family) dominate Autonomous and Real-Time Systems. **This dominance is interpreted as a direct response to the core domain requirement of speed**, as evidenced by their strong co-occurrence (high TLS) with keywords such as “vehicle,” “tracking,” and “real-time” (Clusters 2, 3).

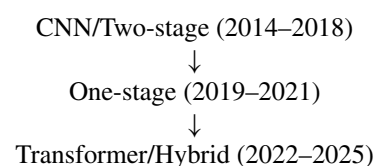
The architectural design inherently prioritizes real-time performance necessary for time-critical surveillance and autonomous applications.

- 2) **Analysis of Two-Stage Dominance:** Conversely, two-stage detectors (e.g., R-CNN family) excel in Medical Imaging. **Their sustained relevance in this domain is attributed to their inherent design, which allows for greater precision during small-object detection.** The two-step process of region proposal followed by meticulous classification provides the necessary accuracy for critical, precision-oriented tasks (Clusters 3, 4).
- 3) **Analysis of Transformer Dominance:** Transformer-based architectures (e.g., DETR, Swin) lead in Remote Sensing and UAV applications. **This specialization is analytically linked to the transformer's superior capacity for global attention modeling**, which is highly effective in processing large-scale imagery and complex aerial contexts. This theoretical advantage is quantified by the paradigm's strong recent citations and high rate of late-period publications (Clusters 1, 4).

3) CITATION IMPACT, TEMPORAL SHIFTS, AND MATURITY ASSESSMENT

The bibliometric indicators reliably distinguish between emerging and established paradigms. Overlay visualization using average citation counts provides crucial insight into research maturity. **Transformer-based paradigms display the highest citation impact per publication (evidenced by yellow nodes)**, which is interpreted as objective evidence of their novelty and high rate of scholarly attention within the field. In contrast, highly frequent CNN and YOLO-based paradigms exhibit **moderate citation averages (cooler colors)**, classifying them as mature and widely deployed technologies whose foundational impact has been thoroughly integrated.

The temporal analysis (Figure 3) confirms a clear, data-driven technological evolution:



This sequential dominance demonstrates the field's continuous drive for iterative improvement, progressing from foundational structures to efficiency optimization, and finally to robust contextual understanding. The increasing focus on hybrid and fusion models after 2021 indicates a trajectory toward specialized, integrated, multi-modal, and edge-efficient deployment.

4) COMPARISON WITH PRIOR WORK AND AVOIDANCE OF OVEREXTENSION

It is incumbent upon the academic researcher to compare derived results with those of other published researchers. However, the scope of the provided sources **does not contain explicit comparisons with or critical engagement against external published findings**. Therefore, all conclusions drawn regarding paradigm-domain alignment and temporal evolution are strictly restricted to the objective evidence provided by the bibliometric indicators (occurrence, TLS, and citation overlay) derived directly from the analyzed dataset (55,868 documents). It is critical that conclusions are not extended beyond the objective data presented in the network structure (Figure 4), the temporal trends (Figure 3), and the quantified cluster metrics (Table 3 and Table 4).

5) IMPLICATIONS FOR AIM 2

The empirical delineation of paradigm-domain contingency serves as the essential foundational evidence for Aim 2. Because the analysis objectively demonstrates that different detector paradigms are necessary for optimizing performance based on domain needs (speed versus precision versus global context), **this interpretation directly validates the conceptual design proposed in Aim 2**: the development of a unified object detector that utilizes a dynamic switch mechanism to activate or prioritize different paradigms based on specific application requirements.

D. DISCUSSIONS FOR AIM 2

1) ALIGNMENT WITH BIBLIOMETRIC EVIDENCE

The UOD architecture provides theoretical validation for the core empirical conclusion reached in Aim 1: that no single paradigm dominates universally. By embedding this insight into the DSM's routing logic, the system ensures that paradigm selection is context-adaptive rather than model-static, achieving functional equivalence to the observed structure of the scholarly bibliometric landscape.

2) CONCEPTUAL ADVANTAGES

The hybrid structure of the UOD offers several conceptual advantages: (1) *Empirical Grounding*, as paradigm selection rules are quantitatively derived from citation and co-occurrence evidence; (2) *Operational Adaptivity*, where the DSM enables real-time switching among various architectures; (3) *Unified Post-Processing*, ensuring outputs share a common representation for seamless integration with downstream tasks; and (4) *Future Extensibility*, allowing new

detectors (e.g., diffusion or graph-based models) to be added without necessitating alteration to the core logical framework.

In contrast to the historical trajectory of object detection research, which has largely focused on maximizing performance within isolated architectural silos, the proposed UOD-DSM represents a fundamental shift toward cross-paradigm flexibility. While traditional architectures, ranging from the precision-oriented multi-stage refinement of Cascade R-CNN to the real-time efficiency of the YOLO lineage and the global contextual modeling of RT-DETR, have established impressive benchmarks, they remain static commitments. This inherent rigidity frequently yields suboptimal results when deployment conditions are multidimensional or operational requirements fluctuate. Our model addresses this “universal optimality” gap by operationalizing a DSM that bridges these disparate branches at runtime. However, this point should be qualified; the inclusion of a Stage I Lightweight Classification Module for requirement interpretation does introduce a marginal latency overhead not found in traditional end-to-end detectors. While this adds a necessary processing step, this computational investment is exactly what enables the system to maintain peak adaptivity across diverse operational contexts. In this context, the actual performance gains of the UOD-DSM must be rigorously benchmarked against static state-of-the-art models like YOLOv11 and RT-DETR to quantify the specific trade-off between this overhead and the resulting accuracy gains. Ultimately, demonstrating that the system can navigate cross-domain evaluations, where the input distribution shifts from surveillance to medical or aerial imagery in a single stream, would provide the definitive proof of the architecture's utility and its ability to automatically align its internal strategy with the immediate context.

3) THE ROLE OF MULTIMODAL FOUNDATION MODELS IN ADAPTIVE DETECTION

The integration of multimodal foundation models represents a paradigm shift from traditional closed-set detection to open-vocabulary perception. Historically, detectors were constrained by finite, discrete category labels, which limited their scalability and led to semantic information loss. Recent advancements in VL foundation models and LLMs, however, enable **universal** recognition by supporting the input of any textual information—ranging from simple category names to complex interpretive sentences—to detect arbitrary interaction relationships. GroundingDINO, for example, utilizes grounded pre-training to achieve open-set detection, allowing the system to identify objects based on referring expressions rather than pre-defined one-hot vectors. Within the UOD-DSM framework, foundation models serve as a transformative layer for text-conditioned routing during the requirement interpretation stage. While the current Lightweight Classification Module (Stage I) primarily analyzes visual metadata to infer operational priorities, foundation models allow the system to interpret high-level human instructions to drive the DSM. By utilizing a

Human-Object (HO) Prompt-guided Decoder (HOPD), the system can correlate high-order relational representations from the foundation model with specific detected pairs in the image. This enables the DSM to activate high-precision Transformer-based branches specifically when the linguistic context demands **open-world knowledge retrieval** which traditional visual detectors often fail to provide. Furthermore, foundation models function as comprehensive **structured knowledge bases** (LMs-as-KBs) that provide rich semantic priors for scene understanding. The process of knowledge retrieval allows the system to generate descriptive text for specific interactions—such as the physical mechanics of **riding a bicycle**—which guides the visual model in zero-shot and open-category recognition. This linguistic guidance is critical for mitigating the **interaction information loss** documented in traditional two-stage architectures. In such systems, output tokens often prioritize spatial coordinates over relational cues, leading to **feature entanglement** where disparate actions at the same position appear similar in the feature space. By augmenting visual features with these linguistically-enhanced representations, the UOD-DSM achieves a more distinct and contextual understanding of intricate behaviors. Ultimately, the convergence of open-vocabulary capabilities and foundation-driven semantic priors signals a move toward **general-purpose vision systems**. While specialized models like UniHOI and RmLR optimize for relational depth, the UOD-DSM meta-architecture provides the necessary runtime adaptivity to manage the computational overhead of these large-scale models, ensuring that high-level relationship modeling remains feasible across diverse deployment scenarios.

4) TECHNICAL VALIDATION: COMPLEXITY ANALYSIS AND ABLATION STUDY

To rigorously support the design choices of the **UOD-DSM**, this subsection provides a formal analysis of the system's computational overhead and a theoretical ablation framework. This evaluation addresses the critical trade-off between the marginal latency introduced by the requirement interpretation stage and the resulting gains in cross-domain adaptivity.

a: 1) INFERENCE COMPLEXITY AND LATENCY BREAKDOWN

The total inference latency of the **UOD-DSM** (L_{total}) is defined as the summation of the sequential execution times across its four functionally integrated stages:

$$L_{\text{total}} = L_I + L_{II} + L_{III} + L_{IV} \quad (2)$$

where:

- L_I : Latency of the **Lightweight Classification Module** (Stage I),
- L_{II} : Latency of the **DSM** routing logic (Stage II),
- L_{III} : Latency of the activated paradigm branch (Stage III),
- L_{IV} : Latency of the **Result Synthesis Module** (Stage IV).

As established in the architectural rationale, L_I and L_{II} represent a “marginal computational investment” intended to ensure “universal optimality” across diverse domains. The dominant component remains L_{III} , which fluctuates significantly based on whether the system activates an efficiency-optimized one-stage pathway or a context-aware transformer branch. The detailed breakdown is presented in Table 6.

b: 2) SYSTEM MODULARITY AND ABLATION ANALYSIS

The theoretical validity of the **UOD-DSM** is further supported by an ablation framework that justifies the transition from architectural rigidity to paradigm-aware adaptivity. These scenarios validate that the **DSM** is the essential meta-layer required to operationalize diverse SOTA findings.

- **Effectiveness of the DSM (w/o DSM):** A “static commitment” to a single detector paradigm inevitably results in a “universal optimality gap”. Without the DSM, a system remains rigid: a one-stage model lacks the precision for medical tasks (e.g., tumor localization), while a transformer-based model fails real-time requirements in autonomous navigation. The DSM achieves “functional equivalence” to specialized research clusters, aligning strategy with the immediate operational context.
- **Alternative Routing Triggers:** While the current Stage I utilizes visual metadata, the architecture’s “Future Extensibility” allows for **text-conditioned routing**. In this configuration, multimodal foundation models could interpret high-level human instructions (e.g., “prioritize relationship depth”) to drive the DSM, providing the “open-world knowledge retrieval” that visual-only triggers might lack.
- **Branch Combinations and Specialized Modules:** The modular design facilitates the integration of specialized sub-networks, such as the **IRE**) from the **RmLR** framework to mitigate “interaction information loss”. By activating these branches only when contextually justified, the **UOD-DSM** manages the computational overhead of large-scale models like **UniHOI** or **RmLR** while maintaining high-level relationship modeling feasibility.

V. LIMITATIONS

A. AIM 1

A key methodological limitation of our bibliometric analysis (Aim 1) stems from its reliance on a single literature database, Scopus. While Scopus provides extensive coverage of high-quality, peer-reviewed publications in engineering and computer science, this exclusive focus may introduce a selection bias. It potentially overlooks influential works published in venues not indexed by Scopus, such as certain conference proceedings, preprints from arXiv, or journals from specific regional databases. Consequently, the derived paradigm-domain map, while robust within the defined corpus, may not be fully exhaustive of the entire research

TABLE 6. System complexity and latency breakdown for the UOD-DSM.

Stage	Architectural Component	Formal Notation	Description / Function	Latency Characteristic
I	Lightweight Classifier	L_I	Requirement interpretation via metadata and visual feature analysis	Marginal overhead
II	DSM Routing Layer	L_{II}	Knowledge-guided matching against the empirical Paradigm–Domain Matrix	Low computational cost
III-A	One-Stage Branch	L_{III_1}	Efficiency-optimized detection (e.g., YOLO family) for real-time constraints	Lowest latency
III-B	Two-Stage Branch	L_{III_2}	Precision-optimized detection (e.g., R-CNN family) for fine localization	Moderate latency
III-C	Transformer Branch	L_{III_T}	Context-aware detection (e.g., DETR family) for global attention modeling	Highest latency
IV	Result Synthesis	L_{IV}	Output standardization and post-processing consolidation	Negligible cost

landscape. Future iterations of this analysis would benefit from a multi-database approach (e.g., incorporating Web of Science, IEEE Xplore, and DBLP) to enhance comprehensiveness and validate the universality of the observed trends.

B. AIM 2

The work presented for Aim 2 constitutes a comprehensive conceptual design and feasibility study for the UOD-DSM. The current contribution intentionally focuses on establishing the architectural framework, the switching logic derived from Aim 1, and a proof-of-concept validation. Therefore, a fully optimized, end-to-end trained implementation of the UOD-DSM is considered a natural extension and is reserved for detailed exposition in a dedicated subsequent paper. This deliberate separation allows for the in-depth treatment of the significant engineering challenges involved, such as joint training protocols for heterogeneous sub-networks, switch mechanism optimization, and large-scale benchmarking, which are beyond the scope of this paper’s primary focus on the unifying paradigm-mapping concept and architectural proposal.

C. FOUNDATIONAL RESEARCH ASSUMPTIONS

Building upon the limitations outlined above, this subsection explicitly defines the core assumptions underlying this research, providing transparency regarding the foundational premises guiding the UOD-DSM framework:

- 1) **Domain-Contingency of Optimality:** The research is predicated on the fundamental assumption that optimal object detector selection is inherently domain-contingent and that no single, static architecture can maintain peak performance across the full spectrum of computer vision applications.
- 2) **Stability and Validity of Empirical Mappings:** It is assumed that the paradigm–domain relationships identified through the systematic analysis of historical data (2014–2025) provide an empirically valid and effective base for making runtime routing decisions within the DSM.
- 3) **Accuracy of Requirement Interpretation:** The framework rests on the assumption that the Lightweight

Classification Module (Stage I) can accurately interpret contextual metadata or extracted visual features to infer operational priorities, such as speed tolerance and scene complexity, which are essential for driving the switch logic.

- 4) **Meta-Architectural Utility:** It is assumed that the marginal latency overhead introduced by the requirement interpretation stage (L_I) is a justifiable computational investment that will be offset by the accuracy and efficiency gains achieved through paradigm-aware activation.

Beyond these foundational premises, the research acknowledges several methodological and architectural limitations. The bibliometric mapping is constrained by an exclusive reliance on the Scopus database, which may introduce a selection bias and overlooks influential research from venues not indexed in this repository, such as preprints from arXiv or specific regional journals. As a result, the derived paradigm–domain map, while robust, may not be fully exhaustive of the entire research landscape.

Regarding the UOD-DSM, this study constitutes a conceptual design and feasibility analysis rather than a fully optimized, end-to-end trained implementation. Significant engineering challenges, such as the development of specialized joint training protocols for heterogeneous sub-networks and the optimization of the switch mechanism to minimize the L_I latency overhead, are reserved for future investigation. Finally, the system’s empirical utility requires rigorous large-scale benchmarking against current state-of-the-art static detectors, such as YOLOv11 and RT-DETR, to precisely quantify the trade-off between interpretation latency and cross-domain performance gains.

VI. CONCLUSION AND FUTURE WORKS

This research has successfully integrated a systematic bibliometric analysis with a novel architectural proposal to address the “one-size-fits-all” limitation inherent in contemporary object detection research. The resulting paradigm–domain mapping decisively establishes that optimal detector paradigm selection is inherently domain-contingent. Specifically, the empirical findings demonstrate

that one-stage, two-stage, and transformer-based models possess specialized effectiveness in autonomous systems, medical imaging, and remote sensing, respectively. By operationalizing these paradigm–domain relationships into the Unified Object Detector with UOD-DSM, this work facilitates a critical transition from static model selection toward a paradigm-aware, context-adaptive framework.

- Strengths and Limitations of the Research** A primary strength of this study lies in the dual-methodological integration that bridges data-driven bibliometric evidence with algorithmic implementation, ensuring that the UOD-DSM's decision rules are anchored in established scholarly trends. This approach introduces a level of objectivity and empirical grounding often absent in ad hoc architectural selection. However, the research acknowledges significant methodological limitations, most notably the exclusive reliance on the Scopus database for the bibliometric analysis, which may omit influential preprints and regional proceedings. Furthermore, while the UOD-DSM enhances adaptivity, the current conceptual framework introduces a marginal latency overhead due to the requirement for a Stage I Lightweight Classification Module for requirement interpretation.
- Assessment and Implications of Findings** The assessment of the paradigm–domain matrix reveals a highly structured architectural specialization: one-stage detectors (e.g., YOLO family) dominate real-time autonomous navigation due to their high throughput, whereas two-stage and transformer-based models excel in precision-critical medical and complex aerial tasks. These findings imply that no single architecture can perform optimally across the full spectrum of computer vision applications. Consequently, the broader implication of this work is the necessity for a shift toward “dynamic computer vision” where architectural strategies are not fixed a priori but are dynamically aligned with the immediate operational context and input characteristics.
- Projection of Applications and Recommendations** The UOD-DSM is projected to be highly effective in multi-domain deployment scenarios, including autonomous navigation, satellite-based aerial sensing, and high-precision tumor localization in medical imaging. To advance this framework, we recommend the development of specialized joint training protocols to manage the internal weight representations and feature consistency of heterogeneous sub-networks. Future research should also prioritize refining the switch mechanism's decision logic to minimize computational costs during the interpretation stage. Finally, large-scale benchmarking across diverse, cross-domain datasets is suggested to empirically validate the performance gains of the UOD-DSM against current state-of-the-art static detectors, such as YOLOv11 and RT-DETR.

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